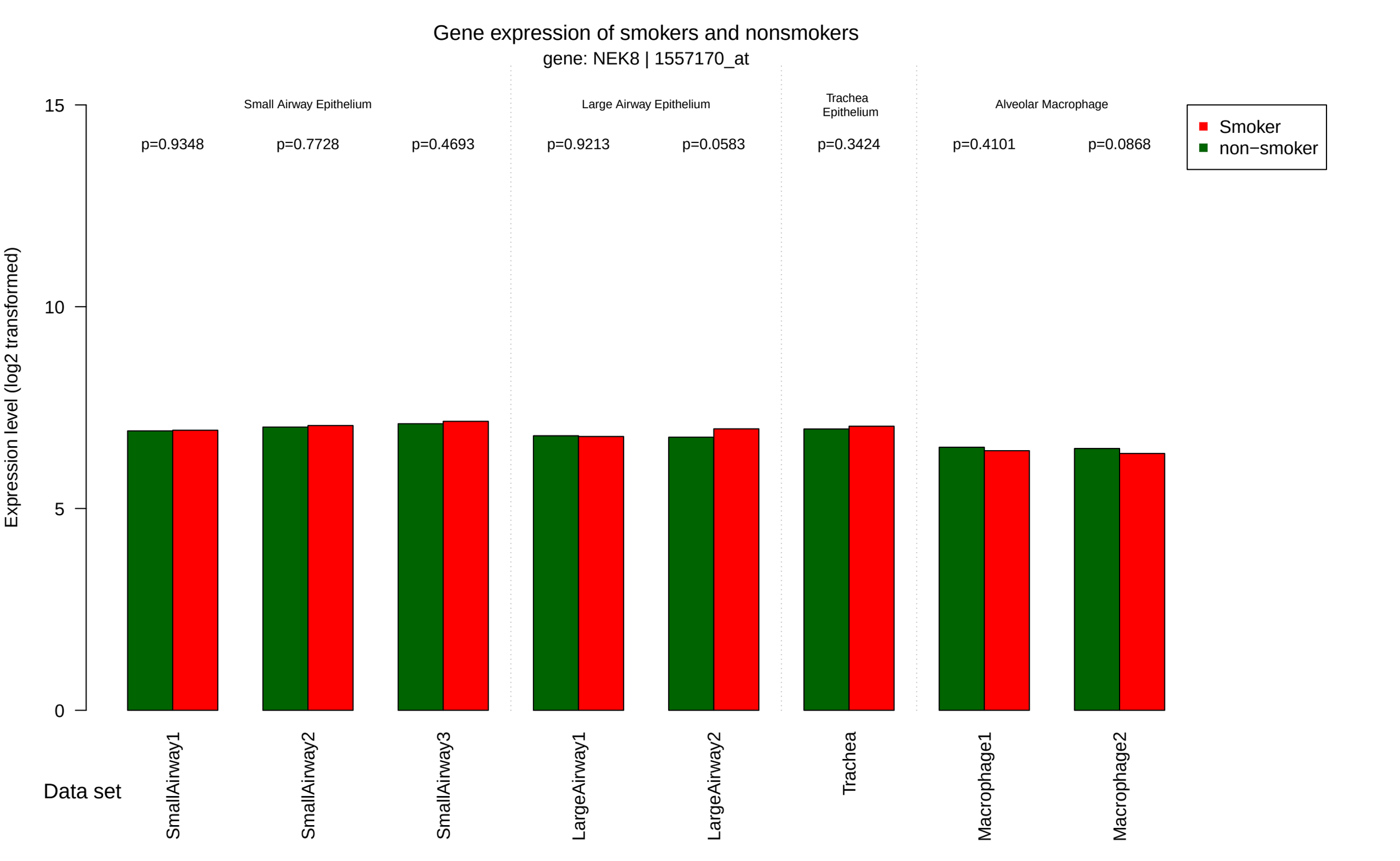
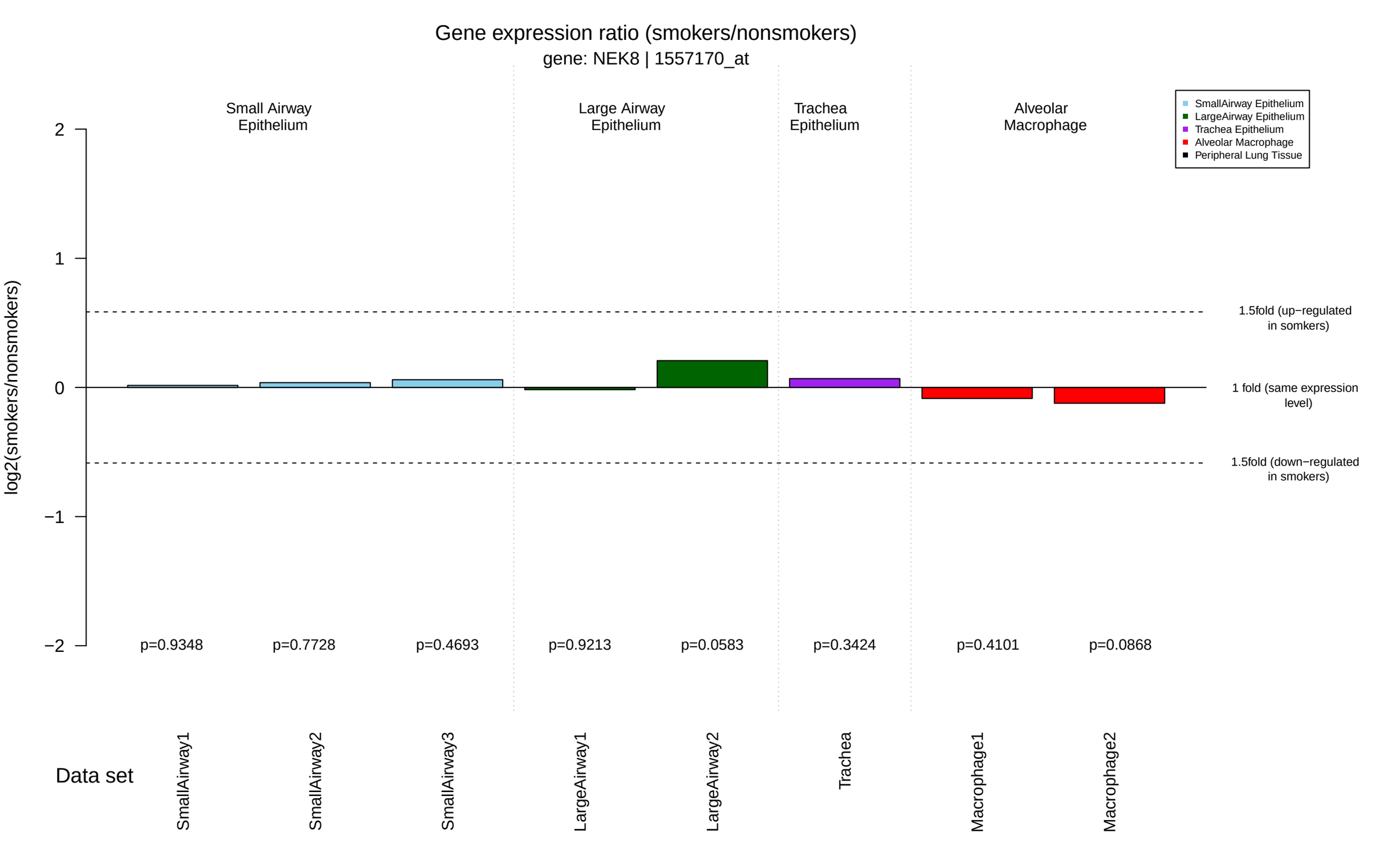
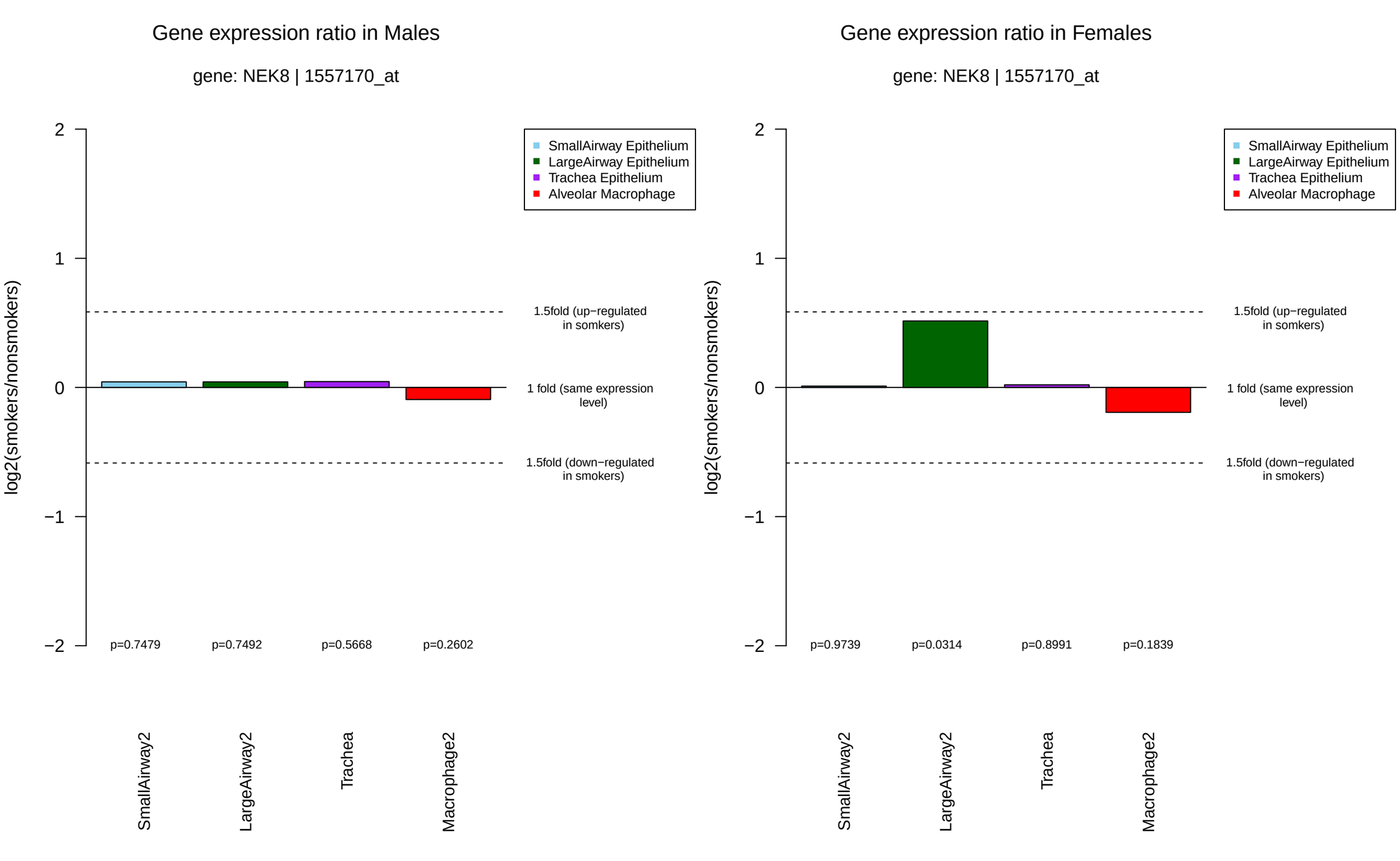
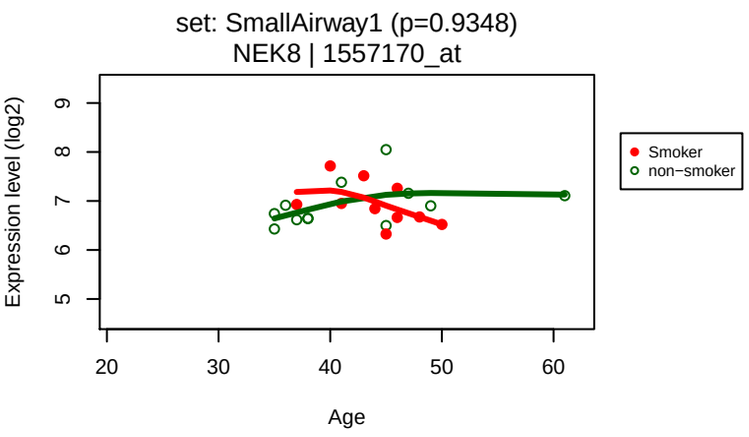
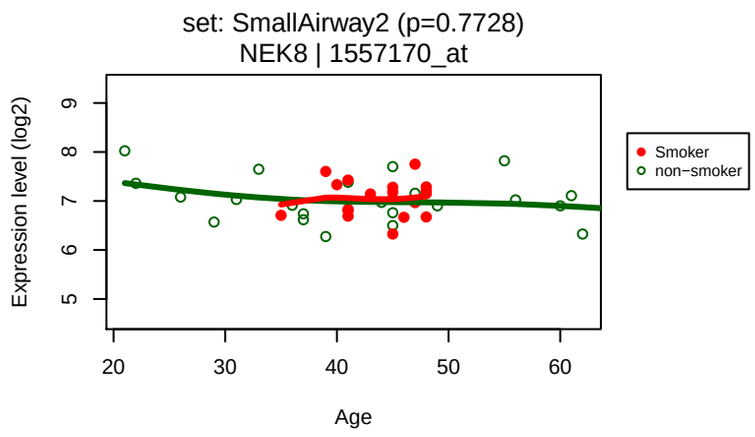
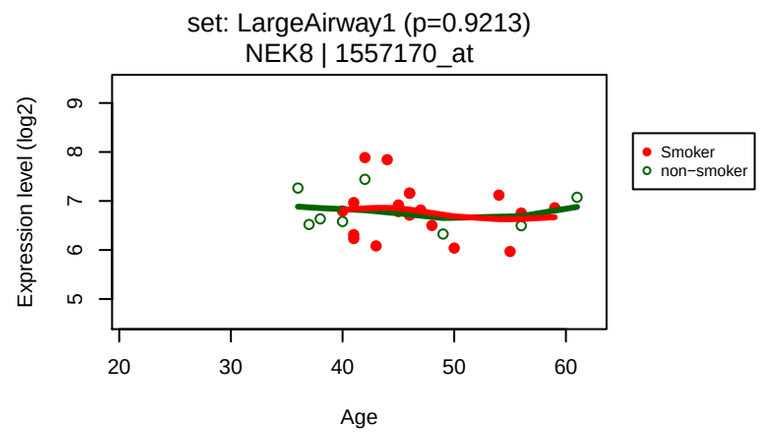

Gene expression of smokers and nonsmokers
gene: NEK8 | 1557170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9348
p=0.7728
p=0.4693
p=0.9213
p=0.0583
p=0.3424
p=0.4101
p=0.0868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

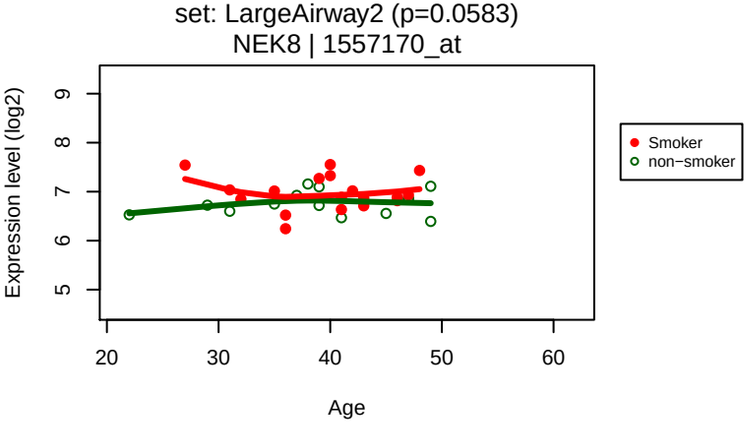

Gene expression ratio (smokers/nonsmokers)
gene: NEK8 | 1557170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9348
p=0.7728
p=0.4693
p=0.9213
p=0.0583
p=0.3424
p=0.4101
p=0.0868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

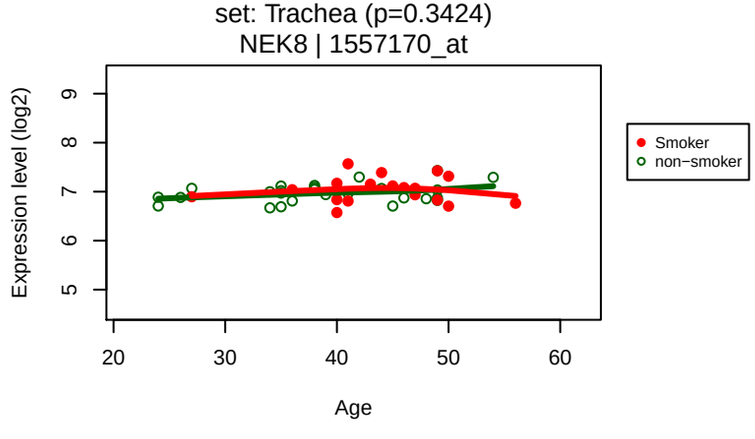

Gene expression ratio in Males
gene: NEK8 | 1557170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7479
p=0.7492
p=0.5668
p=0.2602
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK8 | 1557170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9739
p=0.0314
p=0.8991
p=0.1839
SmallAirway2
LargeAirway2
Trachea
Macrophage2

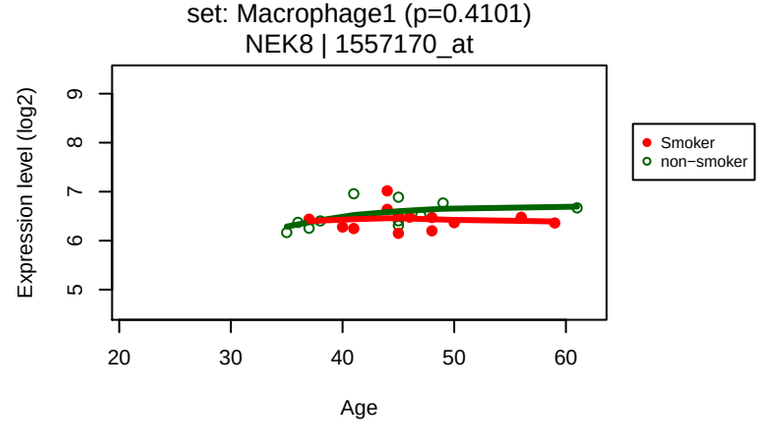

set: SmallAirway1 (p=0.9348)
NEK8 | 1557170_at
Expression level (log2)
Age
Smoker
non−smoker

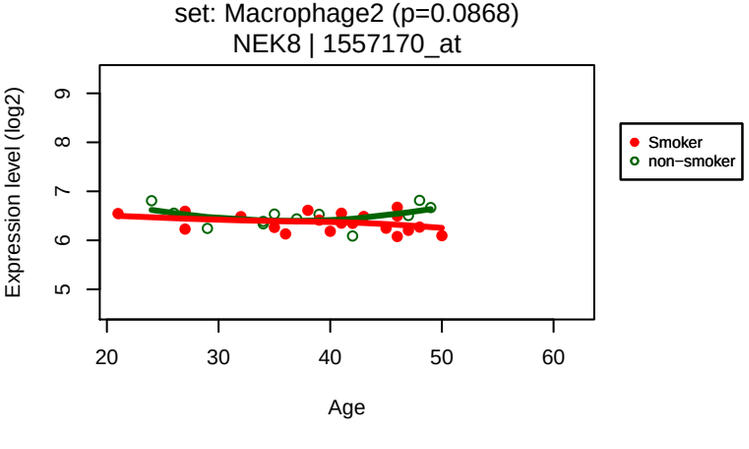

set: SmallAirway2 (p=0.7728)
NEK8 | 1557170_at
Expression level (log2)
Age
Smoker
non−smoker

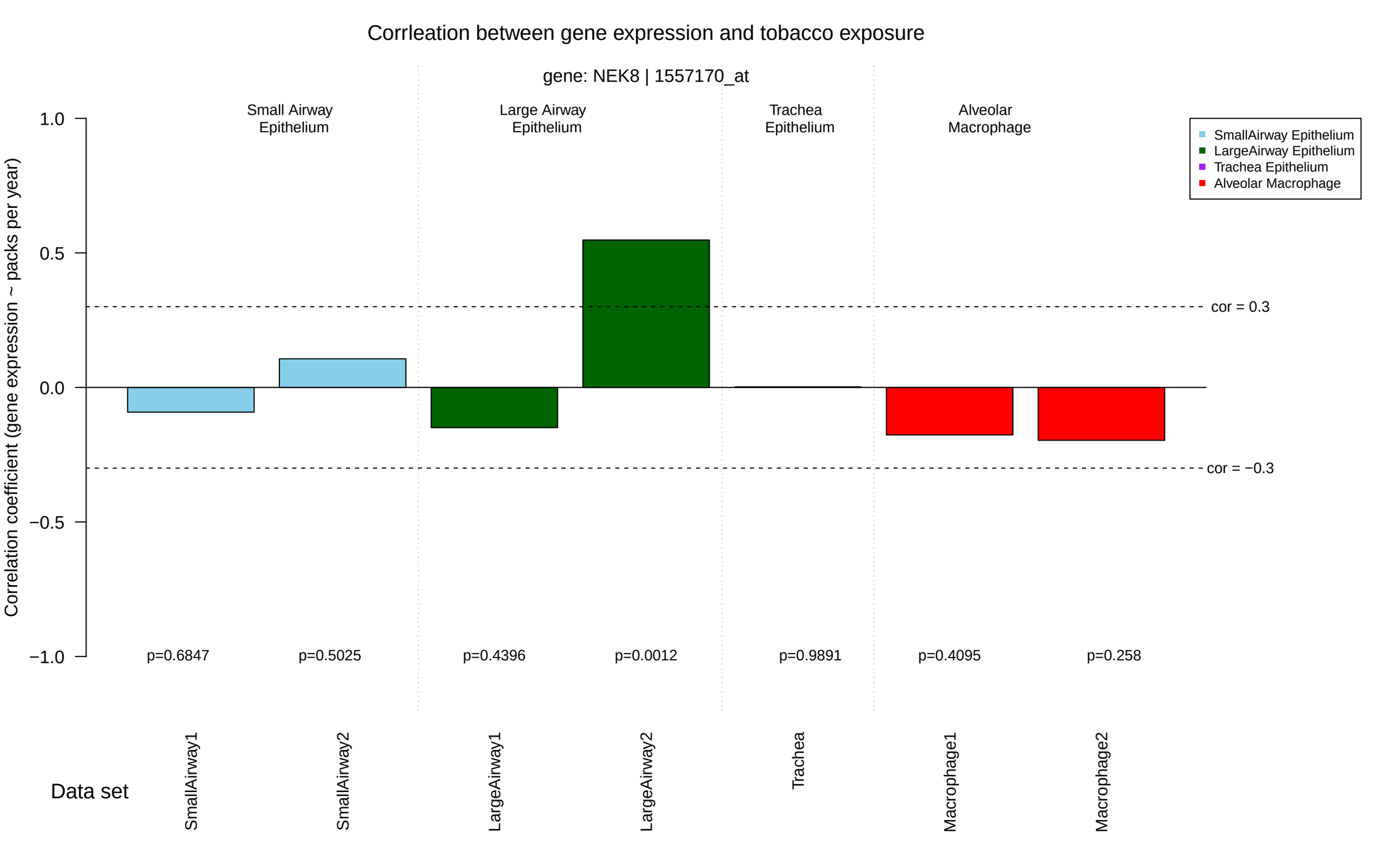

set: LargeAirway1 (p=0.9213)
NEK8 | 1557170_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0583)
NEK8 | 1557170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3424)
NEK8 | 1557170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4101)
NEK8 | 1557170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0868)
NEK8 | 1557170_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK8 | 1557170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6847
p=0.5025
p=0.4396
p=0.0012
p=0.9891
p=0.4095
p=0.258
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2